Signals and system

Answers to some of the internal questions

Typst Source

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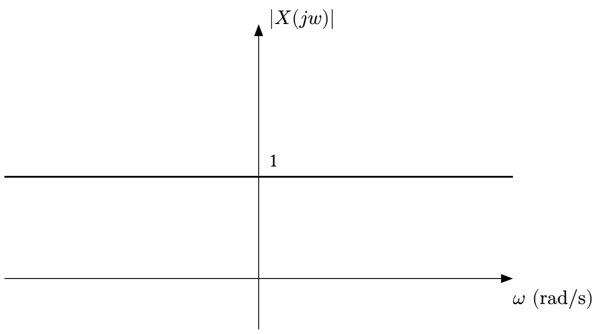
1. Plot the spectrum of the following signals:

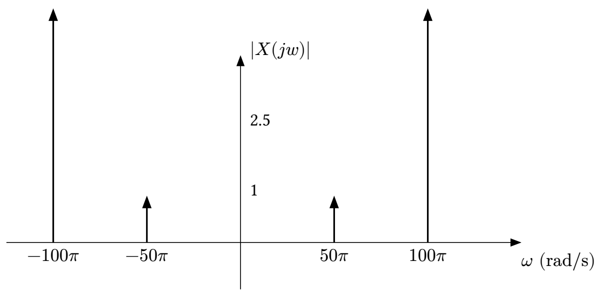
- $x(t) = \delta(t+2)$
- $x(t) = 2\sin(50\pi n) + 5\cos(100\pi n)$
- $x(n) = e^{j20\pi n}$
- $x(t) = e^{j20\pi t}$

Answer:

$x(t) = \delta(t+2)$



$x(t) = 2\sin(50\pi n) + 5\cos(100\pi n)$

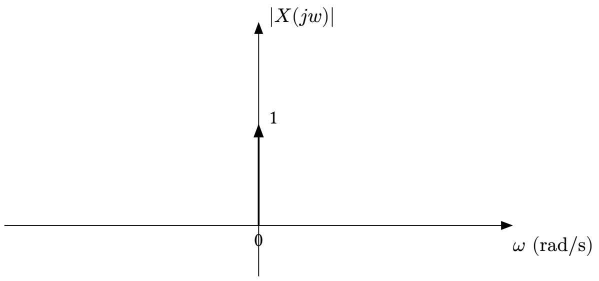


$x(n) = e^{j20\pi n}$

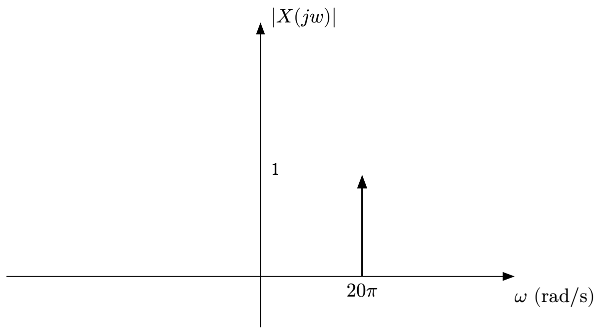
$x(n) = e^{j20\pi n} = \left(e^{j2\pi n}\right)^{10} = 1^{10} = 1$

(since n is an integer, $e^{2\pi jn} = 1$)

This is a DC value



$x(t) = e^{j20\pi t}$



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2. Find the sampling frequency of the signal both the continuous time and the discrete time signal.

$$x(t) = \sin(50\pi t)\cos(100\pi t)$$

Answer:

In continuous time:

$$
\begin{aligned}
x(t) &= \sin(50\pi t)\cos(100\pi t) \\
&= \frac{1}{2}(\sin(50\pi t + 100\pi t) - \sin(50\pi t - 100\pi t) \\
&= \frac{1}{2}(\sin(150\pi t) + \sin(50\pi t)
\end{aligned}
$$

Here maximum frequency is 150π rad/s. Hence sampling frequency should be 300π rad/s

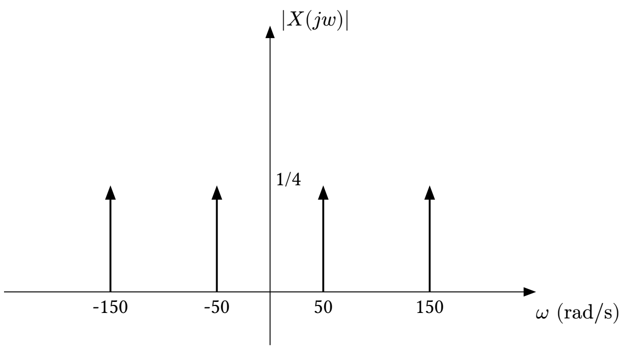
In discrete time:

$$x[n] = \frac{1}{2}(\sin[150\pi n] + \sin[50\pi n]$$

since n is integer, sin of every even multiple of 2π is 0

$$
\begin{aligned}
x[n] &= \frac{1}{2}(\sin[150\pi n] + \sin[50\pi n] \\
&= \frac{1}{2}(\sin[75 \times 2\pi n] + \sin[25 \times 2\pi n] \\
&= 0
\end{aligned}
$$

continuous time spectrum:



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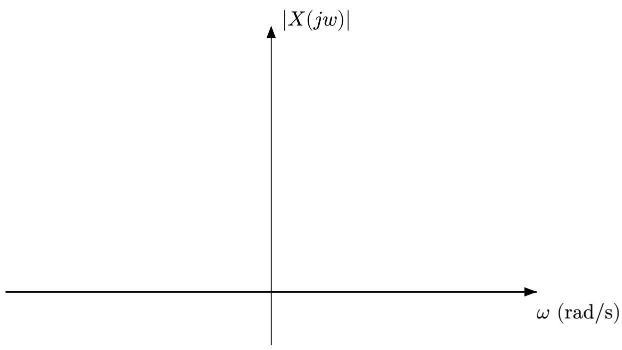
discrete time spectrum:



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3. Find inverse Z transform of

$$H(z) = \frac{1 - \frac{1}{2}z^{-2}}{(1 - z^{-1})(1 + 2z^{-1})}$$

given $|z| > 2$. Comment on stability

Answer:

-> Convert to positive powers of z:

$$
\begin{aligned}
H(z) &= \frac{1 - \frac{1}{2}z^{-2}}{(1 - z^{-1})(1 + 2z^{-1})} \\
&= \frac{z^2 - 0.5}{(z - 1)(z + 2)}
\end{aligned}
$$

Divide by z for partial fraction expansion:

$$\frac{H(z)}{z} = \frac{z^2 - 0.5}{z(z-1)(z+2)} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{z+2}$$

Solve for coefficients:

$$A = \left(\frac{z^2 - 0.5}{(z-1)(z+2)} \right) \Bigg|_{z=0} = \frac{1}{4}$$

$$B = \left(\frac{z^2 - 0.5}{z(z+2)} \right) \Bigg|_{z=1} = \frac{1}{6}$$

$$C = \left(\frac{z^2 - 0.5}{z(z-1)} \right) \Bigg|_{z=-2} = \frac{7}{12}$$

From this :

$$\begin{aligned} H(z) &= \frac{1}{4} + \frac{1}{6}\left(\frac{z}{z-1}\right) + \frac{7}{12}\left(\frac{z}{z+2}\right) \\ &= \frac{1}{4} + \frac{1}{6}\left(\frac{1}{1-z^{-1}}\right) + \frac{7}{12}\left(\frac{1}{1+2z^{-1}}\right) \end{aligned}$$

Given the ROC $|z| > 2$, the sequence is entirely causal:

$$h(n) = \frac{1}{4}\delta(n) + \frac{1}{6}u(n) + \frac{7}{12}(-2)^n u(n)$$

For a causal system to be stable all poles must lie inside the unit circle (their magnitude must be $|z| < 1$). Or the ROC should contain unit circle. Since one of the pole is on −2, (outside of $|z| = 1$) the system is unstable.

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4. The difference equation of LTI is given by:

$$y(n) + 2y(n-1) + 2y(n-2) = x(n)$$

What is output when $x(n) = u(n)$

Answer:

Apply the Z-transform to both sides of the difference equation, assuming zero initial conditions:

$$Y(z) + 2z^{-1}Y(z) + 2z^{-2}Y(z) = X(z)$$

Factor out $Y(z)$ to find the transfer function $H(z)$:

$$\begin{aligned} X(z) &= Y(z)(1 + 2z^{-1} + 2z^{-2}) \\ H(z) &= \frac{Y(z)}{X(z)} \\ &= \frac{1}{1 + 2z^{-1} + 2z^{-2}} \end{aligned}$$

To get output for given input, substitute Z-transform of the unit step input, $X(z) = \frac{1}{1-z^{-1}}$, and convert to positive powers of z:

$$
\begin{aligned}
Y(z) &= \left(\frac{1}{1+2z^{-1}+2z^{-2}}\right)\left(\frac{1}{1-z^{-1}}\right) \\
&= \left(\frac{z^2}{z^2+2z+2}\right)\left(\frac{z}{z-1}\right) \\
&= \frac{z^3}{(z-1)(z^2+2z+2)}
\end{aligned}
$$

Divide by z and factor quadratic term to get partial fraction expansion:

$$
\begin{aligned}
\frac{Y(z)}{z} &= \frac{z^2}{(z-1)(z^2+2z+2)} \\
&= \frac{z^2}{(z-1)(s+1-j)(s+1+j)} \\
&= \frac{z^2}{(z-1)(s-p_1)(s-p_2)} \\
&= \frac{K_0}{z-1} + \frac{K_1}{z-p_1} + \frac{K_2}{z-p_2}
\end{aligned}
$$

Where:

$$
p_1 = -1+j, \quad p_2 = -1+-j
$$

in polar form:

$$
p_1 = \sqrt{2}\exp\left(j\frac{3\pi}{4}\right), \quad p_2 = \sqrt{2}\exp\left(-j\frac{3\pi}{4}\right)
$$

Solve for Coefficients:

$$
\begin{aligned}
K_0 &= \left(\frac{z^2}{(z-p_1)(z-p_2)}\right)\Bigg|_{z=1} = \frac{1}{5} \\
K_1 &= \left(\frac{z^2}{(z-1)(z-p_2)}\right)\Bigg|_{z=-1+j} \\
&= \frac{-2j}{-4j-2} = \frac{j}{1+2j} \\
&= \frac{2}{5} + \frac{1}{5}j \\
K_2 &= \left(\frac{z^2}{(z-1)(z-p_2)}\right)\Bigg|_{z=-1-j} \\
&= \frac{j}{1+2j} \\
&= \frac{2}{5} - \frac{1}{5}j
\end{aligned}
$$

From these,

$$
\begin{aligned}
Y(z) &= K_0\frac{z}{z-1} + K_1\frac{z}{z-p_1} + K_2\frac{z}{z-p_2} \\
Y(z) &= K_0\frac{1}{1-z^{-1}} + K_1\frac{1}{1-p_1z^{-1}} + K_2\frac{1}{1-p_2z^{-1}}
\end{aligned}
$$

Apply the inverse Z-transform rule $a^n u(n)$:

$$y(n) = K_0 u[n] + K_1 (p_1)^n u[n] + K_2 (p_2)^n u[n]$$
$$y(n) = \left[\frac{1}{5} + \left(\frac{2}{5} + j\left(\frac{1}{5}\right)\right)(p_1)^n + \left(\frac{2}{5} - j\left(\frac{1}{5}\right)\right)(p_2)^n\right]u(n)$$

Substitute the polar forms of p_1, p_2:

$$p_1 = \sqrt{2}\exp\left(j\frac{3\pi}{4}\right), \quad p_2 = \sqrt{2}\exp\left(-j\frac{3\pi}{4}\right)$$

$$y(n) = \left[\frac{1}{5} + \left(\frac{2}{5} + \frac{1}{5}j\right)\left(\sqrt{2}\exp\left(j\frac{3\pi}{4}\right)\right)^n + \left(\frac{2}{5} - \frac{1}{5}j\right)\left(\sqrt{2}\exp\left(-j\frac{3\pi}{4}\right)\right)^n\right]u(n)$$

factoring $\left(\sqrt{2}\right)^n$ and expanding inner bracket:

$$y(n) = u(n)\left[\frac{1}{5} + \left(\sqrt{2}\right)^n \left[\frac{2}{5}\exp\left(j\frac{3\pi}{4}n\right) + \frac{1}{5}j\exp\left(j\frac{3\pi}{4}n\right) + \frac{2}{5}\exp\left(-j\frac{3\pi}{4}n\right) - \frac{1}{5}j\exp\left(-j\frac{3\pi}{4}n\right)\right]\right]$$

Here term in blue can be simplfied as:

$$\frac{2}{5}\exp\left(j\frac{3\pi}{4}n\right) + \frac{1}{5}j\exp\left(j\frac{3\pi}{4}n\right) + \frac{2}{5}\exp\left(-j\frac{3\pi}{4}n\right) - \frac{1}{5}j\exp\left(-j\frac{3\pi}{4}n\right)$$
$$= \frac{2}{5}\underbrace{\left[\exp\left(j\frac{3\pi}{4}n\right) + \exp\left(-j\frac{3\pi}{4}n\right)\right]}_{2\cos\left(\frac{3\pi}{4}n\right)} + \frac{1}{5}j\underbrace{\left[\exp\left(j\frac{3\pi}{4}n\right) - \exp\left(-j\frac{3\pi}{4}n\right)\right]}_{-2\sin\left(\frac{3\pi}{4}n\right)}$$
$$= \frac{4}{5}\cos\left(\frac{3}{4}\pi n\right) - \frac{2}{5}\sin\left(\frac{3}{4}\pi n\right)$$

Substituting this in $y(n)$ gives:

$$\boxed{y(n) = \left[\frac{1}{5} + \frac{4}{5}\left(\sqrt{2}\right)^n\cos\left(\frac{3\pi}{4}n\right) - \frac{2}{5}\left(\sqrt{2}\right)^n\sin\left(\frac{3\pi}{4}n\right)\right]u(n)}$$

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5. The Z transform of a input sequence x(n) is given as

$$X(z) = 2z + 4 - 4z^{-1} + 3z^{-2}$$

and if $y(n) = x(n-1)$, find $Y(z)$

Answer:

given $y(n) = x(n-1)$

Using time shifting property: $Y(z) = z^{-1}X(z)$

Substituting given $X(z)$ gives: $Y(z) = z^{-1}(2z + 4 - 4z^{-1} + 3z^{-2})$

$$\boxed{Y(z) = 2 + 4z^{-1} - 4z^{-2} + 3z^{-3}}$$

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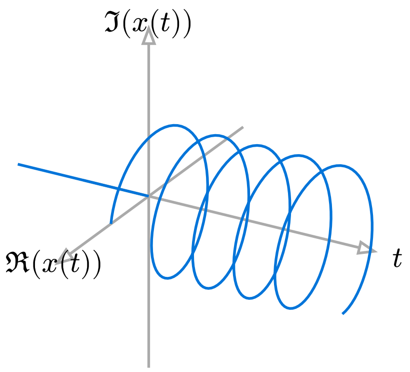
6. Classify signal $x(t) = e^{j10t}u(t)$ into various classes of signals

Answer:



Plot of $x(t) = e^{j10t}u(t)$

Duration: Infinite Duration

The unit step function $u(t)$ turns the signal "on" at $t = 0$. Because the complex exponential e^{j10t} oscillates endlessly as time progresses, the signal extends from $t = 0$ to $t \to \infty$. Therefore, it is an **infinite duration** signal.

Power Signal

Magnitude of the signal is:

$$|x(t)| = |e^{j10t} \cdot u(t)| = |e^{j10t}| \cdot u(t) = u(t)$$

From this energy and power is computed:

- Energy:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt = \int_0^{\infty} 1^2 dt = \infty$$

- Power:

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 \, dt = \lim_{T\to\infty} \frac{1}{2T} \int_0^T 1 dt = \lim_{T\to\infty} \frac{T}{2T} = \frac{1}{2}$$

Because average power is finite and non-zero this is a Power signal.

Periodicity: Aperiodic

For a signal to be periodic, it must satisfy $x(t) = x(t+T)$ for all values of t from $-\infty$ to ∞. The signal is zero for negative time but oscillating for positive time, meaning it does not repeat its past behavior. Therefore, it is **aperiodic**.

Symmetry: Neither Even nor Odd
Because $x(t)$ is entirely zero for $t < 0$ and non-zero for $t > 0$, it is asymmetrical around the y-axis. It is **neither even nor odd**.

Causal
A system or signal is strictly causal if it equals zero for all negative time ($x(t) = 0$ for $t < 0$). Because the entire expression is multiplied by the unit step function $u(t)$ the signal only exists for $t \geq 0$ hence it is a **causal** signal.

Deterministic
A signal is deterministic if its value can be predicted at any given point in time using a mathematical equation. Since given input is a well-defined mathematical function, it is entirely predictable. It is a **deterministic** signal.

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7. Find Z transform of $x(n) = (n-2)0.8^{n-2}u(n-2)$

Answer:

Let $w(n) = n0.8^n u(n)$, hence $x(n) = w(n-2)$

Let $v(n) = 0.8^n u(n)$, hence $w(n) = nv(n)$

Z Z-transform of $x(n)$ is:

$$\begin{aligned} X(z) &= Z[w(n-2)] \\ &= z^{-2} Z[w(n)] \\ &= z^{-2} W(z) \\ W(z) &= Z[nv(n)] \\ &= -z \frac{dZ[v(n)]}{dz} \qquad \text{[Differentiation in Z domain]} \\ &= -z \frac{dV(z)}{dz} \\ V(z) &= Z[0.8^n u(n)] \\ &= \frac{1}{1 - 0.8z^{-1}} \\ &= \frac{z}{z - 0.8} \end{aligned}$$

Hence substituting back:

$$\begin{aligned} W(z) &= -z \frac{dV(z)}{dz} \\ &= -z \frac{d}{dz} \left[\frac{z}{z - 0.8} \right] \\ &= -z \frac{z - 0.8 - z}{(z - 0.8)^2} = -z \frac{-0.8}{(z - 0.8)^2} \\ &= \frac{0.8z}{(z - 0.8)^2} \end{aligned}$$

$$\therefore X(z) = z^{-2}W(z)$$
$$= z^{-2}\frac{0.8z}{(z-0.8)^2}$$
$$= \frac{0.8z^{-1}}{(z-0.8)^2}$$

$$\boxed{X(z) = \frac{0.8z^{-1}}{(z-0.8)^2}}$$

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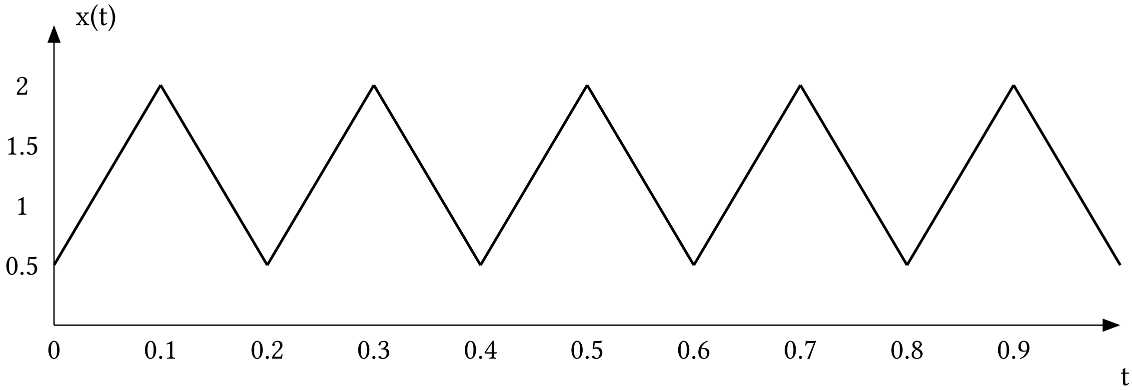
8. Check whether following signal is energy or power signal



Answer:

The signal consists of periodic triangular wave. By definition, all continuous periodic signals are power signals.

Because the wave repeats indefinitely, the total area under its squared magnitude curve is infinite. Therefore, the total energy $E = \infty$.

Because the signal period is finite (between 0.5 and 2), the energy over a single period is finite. Averaging this over time yields a finite average power $0 < P < \infty$.

To calculate power, the piecewise equations for the signal during the first period ($0 \leq t \leq 0.2$) has to be obtained:

$$x(t) = \begin{cases} 15t + 0.5 & \text{for } 0 \leq t \leq 0.1 \\ -15t + 3.5 & \text{for } 0.1 < t \leq 0.2 \end{cases}$$

Thus average power is:

$$P = \frac{1}{T}\int_0^T |x(t)|^2 \, dt$$

Here $T = 0.2$, split the integral for the two segments:

$$P = \frac{1}{0.2}\left[\int_0^{0.1} (15t+0.5)^2 dt + \int_{0.1}^{0.2} (-15t+3.5)^2 dt\right]$$

Evaluate the first integral using the reverse chain rule:

$$\int (at+b)^2 dt = \frac{(at+b)^3}{3a}$$

$$\begin{aligned}\int_0^{0.1} (15t+0.5)^2 dt &= \left[\frac{(15t+0.5)^3}{3(15)}\right]_0^{0.1} \\ &= \frac{(15(0.1)+0.5)^3 - (15(0)+0.5)^3}{45} \\ &= \frac{2^3 - 0.5^3}{45} = \frac{8-0.125}{45} = \frac{7.875}{45} = 0.175\end{aligned}$$

Because the triangle is symmetric, the second integral evaluates to the same area:

$$\int_{0.1}^{0.2} (-15t+3.5)^2 dt = 0.175$$

Final average power:

$$P = \frac{1}{0.2}[0.175+0.175] = 5[0.35] = 1.75$$

Since $P = 1.75$ (a finite, non-zero value) and $E = \infty$, the signal is a power signal.